Milestone 4 - Mechaminds

Here is a summary of our PIC24F16KA301 pinout:

Component	Pic Pin #	Pic pin #	Ccomponent
MCLR	**1**	**20**	3.3 V
A/I (QRD for ball color sensing)	**2**	**19**	GND
A/I (QRD (reading bar codes))	**3**	**18**	A/I (ultrasonic sensor(right))
PWM (left stepper motor driver)	**4**	**17**	A/I (QRD (line detection 1))
PWM (large servo)	**5**	**16**	A/I (QRD (line detection 2))
D/O (left stepper motor driver)	**6**	**15**	A/I (photodiode (line detection 3))
D/O (mosfet for laser)	**7**	**14**	PWM (right stepper motor driver)
A/I (QRD for T junction recognition)	**8**	**13**	D/O (right stepper motor driver)
A/I (IR photodiode laser detection)	**9**	**12**	D/I (IR proximity sensor front)
D/O (mosfet to trigger nerf solenoid)	**10**	**11**	D/I (IR photodiode service station)

List of Components Used

- 1x US -016 Ultrasonic Distance Sensor (Analog)
- 6x QRD Sensor (QRD1114)
- 2x NEMA 17 Stepper Motor (SY42STH38) and Pololu MP6500 Motor Driver and Carrier Board
- 1x Longrunner MG996R Servo
- 2x MOSFET Transistor (IRLB8748PBF)
- 1x 650nm Laser Diode
- 1x IR Proximity Sensor
- 2x IR Photodiode
- 1x LiFePO4 Battery Pack
- 1x 12V Actuated Solenoid
- 1x 5V Buck Converter

Please see pages below for detailed schematic of circuit design:

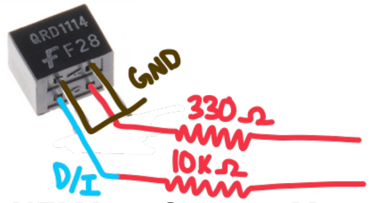
US - 016 Ultrasonic Distance Sensor (Analog)

Vcc 3.3V - 5.0V → Buck converter 12V - 5V

- Max Current 3.8mA

1 A/I Pin [RB15 (#18)]

QRD Sensor (QRD1114)

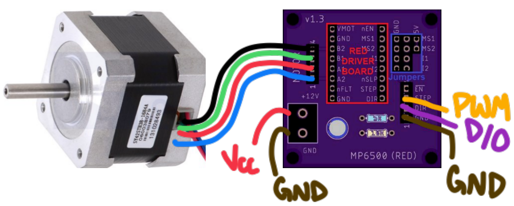


Vcc 3.3V

- Max current 20mA

6 A/I Pins [RB 14 (# 17), RB 13 (# 16), RB 12 (# 15), RA1 (# 3), RA0 (# 2), RA3 (# 8)]

NEMA 17 Stepper Motor (SY42STH38) and Pololu MP6500 Motor Driver and Carrier Board



Vcc 12.0 V

- Max current 1680mA (Carrier board limits this)

2 PWM Pins [RB0 (# 4), RA6 (# 14)]

2 D/O Pins [RB2 (#6), RB9 (#13)]

Longrunner MG996R Servo

Vcc 4.8V - 7.2V → Buck converter 12V - 5V

- Operating current 700mA

1 PWM Pin [RB1 (#5)]

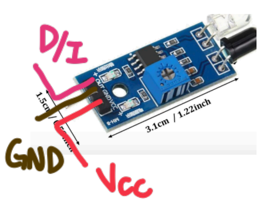
MOSFET Transistor (IRLB8748PBF) and 650nm Laser Diode

Vcc 3.3V

1 D/O Pin [RA2 (#7)]

1 D/O Pin [RA4 (#10)]

IR Proximity Sensor

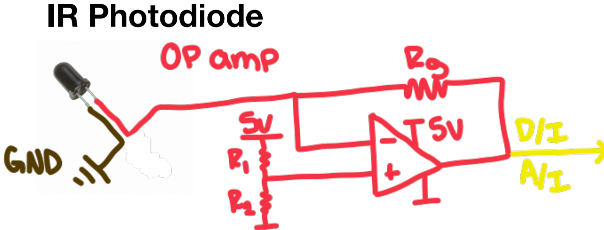


Vcc 3.3V - 5.0V

↳ Buck converter 12V - 5V

1 D/I Pin [RB8 (# 12)]

IR Photodiode



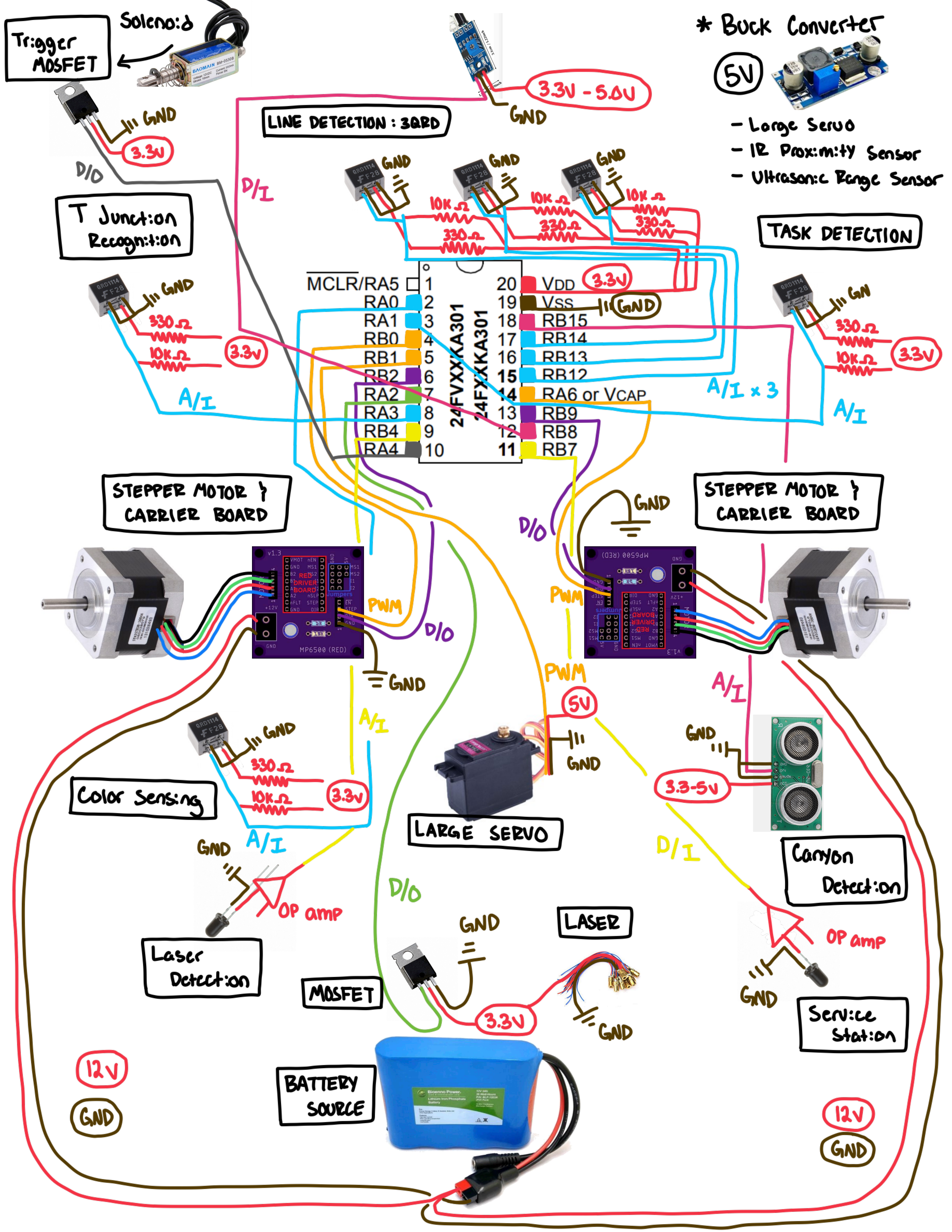
1 A/I Pin [RB4 (#9)]

1 D/I Pin [RB7 (# 11]

LiFePO4 Battery Pack

12V Directly supply motors & PCB to microcontroller

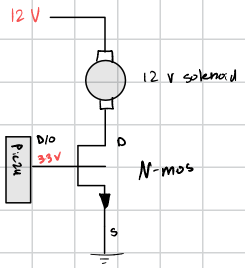
5V Voltage regulator (Buck converter) for {Large Servo, IR Proximity, Ultrasonic Range}



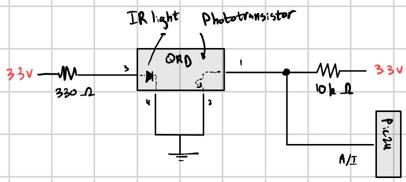
Solenoid
Trigger MOSFET
GND
3.3V
D/O
T Junction Recognition
LINE DETECTION : 3QRD
3.3V - 5.0V
GND
D/I
10K Ω
330 Ω
* Buck Converter
5V
- Large Servo
- IR Proximity Sensor
- Ultrasonic Range Sensor
TASK DETECTION
MCLR/RA5 1
RA0 2
RA1 3
RB0 4
RB1 5
RB2 6
RA2 7
RA3 8
RB4 9
RA4 10
20 VDD
19 VSS
18 RB15
17 RB14
16 RB13
15 RB12
14 RA6 or VCAP
13 RB9
12 RB8
11 RB7
24FVXXKA301
24FXXKA301
3.3V
GND
A/I
A/I x 3
STEPPER MOTOR & CARRIER BOARD
STEPPER MOTOR & CARRIER BOARD
PWM
D/O
GND
PWM
D/O
PWM
5V
GND
A/I
Color Sensing
LARGE SERVO
3.3-5V
A/I
A/I
Canyon Detection
D/I
OP amp
Laser Detection
OP amp
GND
Service Station
D/O
GND
LASER
MOSFET
3.3V
GND
BATTERY SOURCE
12V
GND
12V
GND

Milestone 4 Schematics

Solenoid Trigger Mosfet



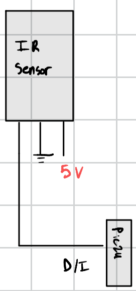
QRD Circuit



Note: - Pins 1-4 are labeled on datasheet.

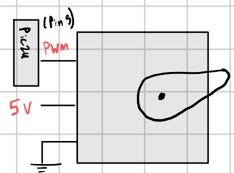
- The dotted lines within the QRD box represents built-in circuitry provided

IR Proximity Sensor



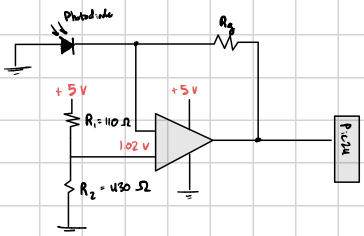
Note: 5 v comes from buck converter

Large Servo



Note: 5 v comes from buck converter

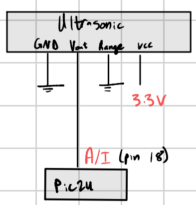
Photodiode



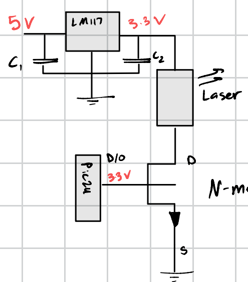
Note: - 5 v comes from buck converter

- R_g will be experimentally picked
- For laser detection, use an A/I pin
- For service station detection, use INTO

Analog Ultrasonic



Laser



Note:

- $C_1 = 10\,\mu F$
- $C_2 = 100\,\mu F$
- Using 3.3 V regulator

Stepper Motors

- As drawn in previous pages

General Notes:

- All 12 v applications come from battery pack
- All 5 v applications come from buck converter
- All 3.3 v applications come from an output pin on Pic24, unless noted otherwise